

AI Understanding

What is Understanding?

- In AI, understanding = ability to:
 1. Interpret input (text, speech, image)
 2. Relate to knowledge base
 3. Produce meaningful response
-  Example: Siri answering 'What is the capital of India?' → New Delhi

Why is Understanding Important?

- • Enables AI to communicate, solve problems, reason
- • Without understanding → only memorization (rote learning)
-  Example: Chatbots giving useful customer support

❁ What Makes Understanding Hard?

- 1. Ambiguity – 'I saw the man with the telescope'
- 2. Context – 'It's cold here'
- 3. Incomplete input – 'Rains tomorrow?'
- 4. Lack of common sense – 'Fire is hot'
- ☞ Example: Google Assistant misinterpreting vague queries

Example Problem

- • Question: 'What is the color of the sky?'
- - Humans → 'Blue (usually)'
- - AI → must check knowledge, context (day/night)
- ☞ Example: Weather app answers differently at night

🌳 Advanced: Constraint Satisfaction

- • Understanding = Constraint Satisfaction Problem (CSP)
- - Solution must satisfy all rules
- ✂ Examples:
 1. 'The bank is near the river' → river bank
 2. 'Who is PM of India?' → Narendra Modi
 3. Grammar parsing as constraints
- ☞ Example: Grammarly checking grammar

Techniques for Understanding

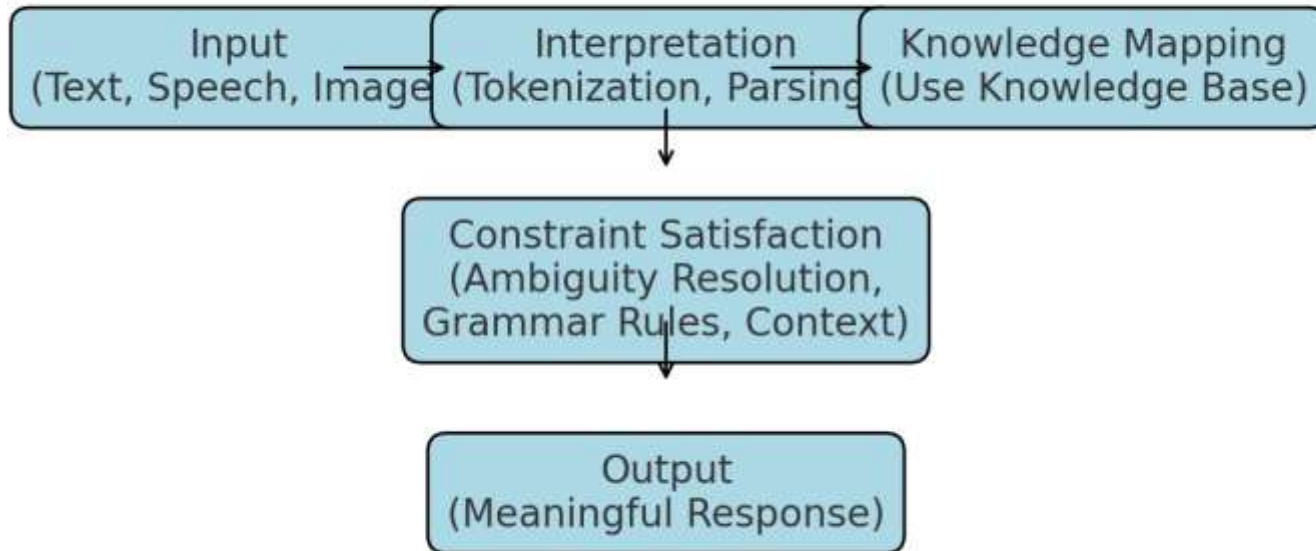
- 1. Logic-Based → First-order logic
- 2. Probabilistic → Handle multiple meanings
- 3. Knowledge Graphs → Facts & relations
-  Example: Google Knowledge Graph for search

✓ Summary Table

- Beginner → Interpret input → Siri answers 'Capital of India'
- Intermediate → Ambiguity/context → 'Man with telescope'
- Advanced → CSP → 'Bank near river' = river bank

AI Understanding Flow Diagram

AI Understanding Flow



IN Real-Life AI Understanding in India (Beginner Level)

- • Voice Assistants in Indian Languages → Google Assistant in Hindi, Tamil, Telugu
 - ☞ Example: 'आज दिल्ली का मौसम कैसा है?'
- • Chatbots in Banking & Railways → SBI YONO, IRCTC AskDISHA
 - ☞ Example: 'Book ticket from Ahmedabad to Mumbai'

IN AI Understanding in India (Intermediate Level)

- • Ambiguity in Indian Languages → 'कल' = yesterday OR tomorrow
- • Education Apps (BYJU's, Vedantu) → Answer questions in English & Hindi
- • Agriculture Apps (Kisan Suvidha) → Farmers ask crop queries in local languages

IN AI Understanding in India (Advanced Level)

- • Healthcare Chatbots (Practo, Apollo 24/7) → Understand symptoms in Hindi
- • Legal Assistants (Indian Kanoon AI) → Answer law-related queries
- • E-Governance (UMANG, DigiLocker) → Understand 'Download Aadhaar card' queries

✓ Indian Summary Table

- Beginner → Chatbots (IRCTC, SBI) → 'Book ticket to Mumbai'
- Intermediate → BYJU's tutor, Kisan apps
- Advanced → Apollo chatbot, Indian Kanoon AI, UMANG governance systems